JUSTIN R. RAMIREZ

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PROFESSIONAL SUMMARY

Software Engineer skilled in Python, Java, C++, and secure application development. Experienced in building reinforcement learning agents, encrypted password managers, and backend systems. Strong foundation in algorithms, system design, secure coding, and full-stack development.

TECHNICAL SKILLS

- **Languages**: Python, Java, C++, SQL, MySQL
- **Tools & Frameworks:** Android Studio, SQLite, MongoDB, Git, GitHub, VS Code, PyCharm, Eclipse, Visual Studio
- **Core Skills:** OOP, CRUD operations, UI/UX fundamentals, data validation, debugging, version control, Agile, secure coding.

PROJECT EXPERIENCE

Maze Intelligence Agent – Python (Reinforcement Learning)

Tech: Python, NumPy, Jupyter, Neural Networks, RL

- Built a pirate-themed deep Q-learning agent that learns to navigate a maze and reach treasure.
- Implemented Q-training loop with ε-greedy exploration, experience replay, and batch updates.
- Tuned hyperparameters and added performance-based early stopping.
- Integrated with provided environment and neural network modules. Impact: Demonstrated practical reinforcement learning, Q-value estimation, and iterative model improvement.

SecureVault – Python, SQLite, AES Encryption

Tech: Python, SQLite, AES, Tkinter

- Developed a secure, multi-user password manager with encrypted credential storage.
- Implemented AES-encrypted vaults, salted authentication, and structured SQLite storage.
- Built a strong password generator and modular Python architecture.
- Designed a simple UI for account creation, login, and credential retrieval. Impact: Reinforced secure coding, encryption workflows, and scalable application design.

Course Manager – C++

Tech: C++, STL, File I/O

- Created a command-line tool to manage and search university course data.
- Added alphabetical sorting, case-insensitive search, and robust input validation.
- Used modular design with helper functions and header files. Impact: Strengthened data structure fluency and backend logic simulation.

EDUCATION

BS – **Computer Science, Southern New Hampshire University**

Graduated • GPA: 3.6 Relevant Coursework: Secure Coding, Full Stack Development, Mobile Architecture, Client/Server Development, Emerging Systems Architecture & Tech

AS – **Computer Science, Southern New Hampshire University**

Graduated Relevant Coursework: Data Structures & Algorithms, Systems Analysis, SDLC, Software Testing, Automation QA

WORK EXPERIENCE

Warehouse Worker – Express Employment | Spokane, WA | 2023–2024

- Improved inventory workflows and reduced fulfillment errors.
- Maintained strict documentation and operated systems with zero safety incidents.
- Collaborated across teams to meet production goals.

Fabricator – Pexco Aerospace | Union Gap, WA | 2017–2019

- Assembled components to engineering specifications with strong quality control.
- Documented inspections and supported engineering teams.
- Maintained a zero-incident safety record.

Machine Operator – Keyes Fiber Corporation | Wenatchee, WA | 2016–2017

- Operated and monitored machinery, troubleshooting issues to reduce downtime.
- Performed quality checks and maintained detailed logs.